

Assignment 1: Bell States and Quantum Teleportation

Title

Generation of Bell States and Simulation of Quantum Teleportation

Aim

To generate Bell states using Hadamard and CNOT gates and to simulate quantum teleportation of an arbitrary qubit using entanglement and classical communication.

Theory

1. Introduction to Qubits

A qubit is the fundamental unit of quantum information. Unlike classical bits, a qubit can exist in a linear combination of states called superposition:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where α and β are complex numbers such that:

$$|\alpha|^2 + |\beta|^2 = 1$$

This property allows quantum computers to process multiple possibilities simultaneously.

2. Quantum Entanglement

Quantum entanglement is a phenomenon where two qubits become interdependent. The state of one qubit cannot be described independently of the other.

If two qubits are entangled, measuring one qubit immediately determines the state of the other, regardless of distance. This non-classical correlation is essential for quantum communication.

3. Bell States

Bell states are maximally entangled two-qubit states. They are defined as:

$$\Phi^+ = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$\Phi^- = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$\Psi^+ = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

$$\Psi^- = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

These states form an orthonormal basis and are widely used in quantum information theory.

4. Generation of Bell States

Steps involved:

1. Initialize two qubits in state $|00\rangle$
2. Apply Hadamard gate to the first qubit $\rightarrow$ creates superposition
3. Apply CNOT gate $\rightarrow$ entangles the qubits

Different Bell states can be generated by applying additional Pauli gates (X, Z).

5. Quantum Teleportation

Quantum teleportation transfers the state of a qubit from one location to another using entanglement and classical communication.

Procedure:

1. Prepare an unknown quantum state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

2. Create an entangled Bell pair between sender (Alice) and receiver (Bob)
3. Alice performs:
 - CNOT operation
 - Hadamard transformation
 - Measurement of her qubits
4. Alice sends 2 classical bits to Bob
5. Bob applies correction operations:
 - 00 $\rightarrow$ Identity
 - 01 $\rightarrow$ X gate
 - 10 $\rightarrow$ Z gate

- $11 \rightarrow X \text{ and } Z$

6. Important Principles

- **No-Cloning Theorem:** Unknown quantum states cannot be copied
- **Measurement Collapse:** Measuring a qubit collapses its state
- **Entanglement Resource:** Required for teleportation

7. Applications

- Secure quantum communication
- Quantum networks
- Quantum cryptography

Conclusion

The experiment successfully demonstrated the generation of Bell states and the simulation of quantum teleportation. It confirmed that quantum information can be transmitted using entanglement and classical communication, without physically sending the particle.

Assignment 2: Multi-Qubit Gates and Entanglement

Title

Implementation of Multi-Qubit Gates and Analysis of Entanglement

Aim

To implement quantum circuits using Hadamard, Pauli-X, CNOT, and Toffoli gates and study their effects on multi-qubit systems.

Theory

1. Quantum Gates

Quantum gates are unitary operations that modify the state of qubits.

2. Important Quantum Gates

Hadamard Gate (H)

- Converts basis states into superposition
- $|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$

Pauli-X Gate (X)

- Acts as a NOT gate
- $|0\rangle \rightarrow |1\rangle$

CNOT Gate

- A two-qubit gate
 - Flips the target qubit if control qubit is 1
 - Used to create entanglement
-

3. Toffoli Gate (CCNOT)

The Toffoli gate is a three-qubit gate with two control qubits and one target qubit.

$$|a, b, c\rangle \rightarrow |a, b, c \oplus (a \cdot b)\rangle$$

This means:

- Target flips only when both controls are 1

4. Properties of Toffoli Gate

- Reversible (no information loss)
- Universal for classical computation
- Used in quantum circuits for complex logic

5. Multi-Qubit Entanglement

When multiple qubits interact through gates like CNOT, they become entangled. This leads to complex correlations that cannot be described independently.

Example:

$$\frac{|000\rangle + |111\rangle}{\sqrt{2}}$$

6. Measurement in Quantum Systems

- Measurement collapses the quantum state
- Results are probabilistic
- Repeated runs provide probability distribution

7. Applications

- Quantum computing algorithms
- Quantum error correction
- Simulation of physical systems

Conclusion

The experiment demonstrated the working of multi-qubit gates and their role in creating entanglement. The Toffoli gate showed how conditional logic is implemented in quantum systems. This provides a foundation for building complex quantum circuits.

Assignment 3: Quantum Fourier Transform (QFT)

Title

Implementation and Verification of Quantum Fourier Transform

Aim

To implement QFT and inverse QFT on multi-qubit systems and analyze the transformation of quantum states.

Theory

1. Fourier Transform Concept

The Fourier Transform converts a signal from time domain to frequency domain. QFT is its quantum equivalent.

2. Mathematical Representation

$$|x\rangle \rightarrow \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{\frac{2\pi i x k}{2^n}} |k\rangle$$

Where:

- n = number of qubits
 - $N = 2^n$
-

3. QFT Circuit Components

- Hadamard gates
 - Controlled phase rotation gates
 - Swap gates (for reversing qubit order)
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4. Inverse QFT

- Reverses QFT operation
- Uses inverse phase rotations
- Recovers original input state

5. Properties of QFT

- Linear transformation
- Reversible
- Efficient (polynomial time complexity)

6. Applications

- Shor's Algorithm (factoring large numbers)
- Quantum phase estimation
- Cryptography
- Signal processing

7. Measurement Analysis

- Output represents frequency components
- Measurement gives probability distribution
- Helps identify periodicity

Conclusion

The Quantum Fourier Transform was successfully implemented and verified using inverse QFT. The experiment demonstrated how quantum systems efficiently perform frequency transformations, which are essential in many advanced quantum algorithms.

Assignment 4: Quantum Measurement and Entanglement

Title

Study of Quantum Measurement, Superposition, and Entanglement using Quantum Circuits

Aim

To construct quantum circuits demonstrating superposition and measurement, implement entanglement between qubits, and analyze the results to understand state collapse and Bell inequality concepts.

Theory

1. Quantum Superposition

A qubit can exist in multiple states simultaneously:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

This property is called **superposition**, and it allows quantum systems to process many possibilities at once.

The **Hadamard gate (H)** is commonly used to create superposition:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

2. Quantum Measurement

Measurement is the process of observing a qubit, which forces it to collapse into one of the basis states.

- Probability of measuring $|0\rangle = |\alpha|^2$
- Probability of measuring $|1\rangle = |\beta|^2$

Once measured, the qubit loses its superposition and remains in the measured state.

3. State Collapse

Before measurement:

- Qubit exists in superposition

After measurement:

- Qubit collapses to a definite state

This is a key difference between classical and quantum systems.

4. Quantum Entanglement

Entanglement is created when two qubits are correlated such that the state of one depends on the other.

Example:

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Measurement of one qubit instantly determines the other.

5. Bell Inequality

Bell's inequality distinguishes between:

- Classical correlations
- Quantum correlations

Quantum systems can **violate Bell inequality**, proving that entanglement cannot be explained by classical physics.

6. Circuit Implementation

- Apply Hadamard → create superposition
 - Apply CNOT → create entanglement
 - Measure qubits → observe correlated outcomes
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7. Applications

- Quantum cryptography
 - Quantum communication
 - Quantum key distribution (QKD)
-

Conclusion

The experiment demonstrated how superposition is created and how measurement collapses quantum states. Entanglement was successfully implemented, and correlated measurement results verified non-classical behavior. This provided a clear understanding of quantum measurement and entanglement principles.

Assignment 5: Deutsch-Jozsa Algorithm

Title

Implementation of Deutsch-Jozsa Algorithm for Determining Constant and Balanced Functions

Aim

To implement the Deutsch-Jozsa algorithm for a 2-qubit system and determine whether a given function is constant or balanced using quantum computation.

Theory

1. Problem Definition

Given a function:

$$f(x): \{0,1\}^n \rightarrow \{0,1\}$$

We need to determine whether the function is:

- **Constant** → same output for all inputs
 - **Balanced** → equal number of 0s and 1s
-

2. Classical vs Quantum Approach

- Classical method requires multiple evaluations
 - Quantum method solves it in **one evaluation** using superposition
-

3. Algorithm Steps

1. Initialize qubits:
 - Input qubits in $|0\rangle$
 - Output qubit in $|1\rangle$
2. Apply Hadamard gates:
 - Creates superposition of all inputs
3. Apply Oracle function U_f :
 - Encodes function behavior
4. Apply Hadamard again

5. Measure input qubits

4. Interpretation of Results

- Output = all zeros → function is **constant**
 - Output ≠ all zeros → function is **balanced**
-

5. Quantum Parallelism

All possible inputs are evaluated simultaneously due to superposition, which gives quantum advantage.

6. Oracle Concept

The oracle is a black-box function that performs:

$$|x, y\rangle \rightarrow |x, y \oplus f(x)\rangle$$

7. Applications

- Foundation for many quantum algorithms
 - Demonstrates quantum speedup
 - Used in complexity theory
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Conclusion

The Deutsch-Jozsa algorithm efficiently determined whether a function is constant or balanced in a single evaluation. This demonstrates the power of quantum parallelism and highlights the advantage of quantum computation over classical methods.

Assignment 6: Shor's Algorithm (Quantum Factorization)

Title

Simulation of Shor's Algorithm for Integer Factorization using Quantum Circuits

Aim

To implement and simulate Shor's algorithm to factor a composite number (e.g., 15) and analyze the role of quantum phase estimation and period finding.

Theory

1. Introduction to Factorization Problem

Factorization involves finding prime factors of a number:

Example:

$$15 = 3 \times 5$$

Classically, this is computationally expensive for large numbers.

2. Shor's Algorithm Overview

Shor's algorithm efficiently factors numbers using:

- Quantum parallelism
 - Quantum Fourier Transform
 - Period finding
-

3. Key Idea: Period Finding

Choose a random number a such that:

$$\gcd(a, N) = 1$$

Find the period r such that:

$$a^r \equiv 1 \pmod{N}$$

4. Quantum Phase Estimation (QPE)

QPE is used to estimate the phase related to periodicity.

Steps:

- Prepare superposition
 - Apply controlled unitary operations
 - Apply inverse QFT
 - Measure to estimate phase
-

5. Role of QFT

QFT extracts periodicity from quantum states, enabling efficient computation of the period.

6. Factor Extraction

Once period r is found:

- If r is even:

$$\gcd(a^{r/2} \pm 1, N)$$

This gives the factors of N .

7. Example (N = 15)

Choose $a = 2$:

- Find period $r = 4$
- Compute:

$$\begin{aligned}\gcd(2^2 - 1, 15) &= 3 \\ \gcd(2^2 + 1, 15) &= 5\end{aligned}$$

8. Applications

- Breaking RSA encryption
- Cryptography
- Number theory

9. Importance

Shor's algorithm demonstrates **exponential speedup** over classical algorithms.

Conclusion

Shor's algorithm successfully demonstrated quantum factorization by finding the period of a function using quantum techniques. The simulation verified the ability to factor small numbers and highlighted the importance of quantum computing in cryptography and computational complexity.

Assignment 7: Quantum Support Vector Machine (QSVM)

Title

Implementation of Quantum Support Vector Machine (QSVM) and Performance Comparison with Classical SVM

Aim

To implement a QSVM using Qiskit's quantum machine learning module, train it on a binary classification dataset, and compare its performance with a classical SVM.

Theory

1. Introduction to Support Vector Machines (SVM)

Support Vector Machines are supervised learning models used for classification. They work by finding an optimal hyperplane that separates data into different classes.

For linearly separable data:

$$w \cdot x + b = 0$$

For non-linear data, kernel functions are used to map data into higher-dimensional space.

2. Kernel Trick

The kernel function allows computation in high-dimensional space without explicitly transforming the data:

$$K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$$

Common kernels:

- Linear
 - Polynomial
 - Radial Basis Function (RBF)
-

3. Quantum Support Vector Machine (QSVM)

QSVM replaces the classical kernel with a **quantum kernel**, computed using quantum circuits.

Steps:

1. Encode classical data into quantum states (feature map)
 2. Compute inner product using quantum circuits
 3. Construct kernel matrix
 4. Train SVM using this kernel
-

4. Quantum Feature Map

A feature map transforms classical data into quantum states:

$$x \rightarrow |\phi(x)\rangle$$

This allows complex patterns to be captured using quantum states.

5. Advantages of QSVM

- Handles high-dimensional data efficiently
 - Captures complex correlations
 - Potential exponential speedup
-

6. Comparison with Classical SVM

Aspect	Classical SVM	QSVM
Kernel	Classical	Quantum
Computation	CPU-based	Quantum circuit
Complexity	Polynomial	Potential speedup

7. Applications

- Image classification
 - Medical diagnosis
 - Fraud detection
-

Conclusion

The QSVM model successfully classified the dataset and demonstrated the use of quantum kernels. The comparison with classical SVM highlighted the potential of quantum computing in handling complex data structures more efficiently.

Assignment 8: Variational Quantum Classifier (VQC)

Title

Implementation of Variational Quantum Classifier using Parameterized Quantum Circuits

Aim

To implement a VQC model, train it on a dataset, and analyze the effect of circuit depth on classification performance.

Theory

1. Variational Quantum Circuits (VQC)

VQC uses parameterized quantum circuits with adjustable parameters (angles) that are optimized during training.

2. Working Principle

1. Encode data into quantum states
 2. Apply parameterized gates
 3. Measure output
 4. Optimize parameters using classical optimizer
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3. Mathematical Representation

A parameterized circuit:

$$|\psi(\theta)\rangle = U(\theta) |0\rangle$$

Where:

- θ are trainable parameters
 - $U(\theta)$ is a unitary operation
-

4. Hybrid Quantum-Classical Model

- Quantum circuit $\rightarrow$ generates output
- Classical optimizer $\rightarrow$ updates parameters

Common optimizers:

- Gradient descent
 - COBYLA
 - SPSA
-

5. Circuit Depth

- Depth = number of layers in circuit
 - Higher depth → more expressive power
 - Too high → noise and overfitting
-

6. Training Process

- Define loss function
 - Update parameters iteratively
 - Minimize classification error
-

7. Applications

- Pattern recognition
 - Classification problems
 - Quantum machine learning
-

Conclusion

The VQC model successfully classified the dataset using parameterized circuits. The study showed that circuit depth plays a critical role in model performance, balancing expressibility and noise.

Assignment 9: BB84 Quantum Cryptography Protocol

Title

Implementation of BB84 Quantum Key Distribution Protocol

Aim

To implement the BB84 protocol for secure key exchange and analyze the impact of eavesdropping on communication.

Theory

1. Quantum Cryptography

Quantum cryptography uses quantum mechanics to ensure secure communication. It is fundamentally secure due to quantum properties.

2. BB84 Protocol Overview

Proposed by Bennett and Brassard (1984), it allows two parties (Alice and Bob) to share a secret key.

3. Encoding Bases

Two bases are used:

- Rectilinear (+): $|0\rangle, |1\rangle$
 - Diagonal ($\times$): $|+\rangle, |-\rangle$
-

4. Steps of BB84

1. Alice sends qubits encoded in random bases
 2. Bob measures using random bases
 3. Compare bases over classical channel
 4. Keep only matching basis results
 5. Generate secret key
-

5. Eavesdropping Detection

If an eavesdropper (Eve) intercepts:

- Measurement disturbs qubits
- Errors increase in key

Error rate indicates presence of attacker.

6. Security Principle

Based on:

- Measurement disturbance
 - No-cloning theorem
-

7. Applications

- Secure communication
 - Quantum internet
 - Military encryption
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Conclusion

The BB84 protocol successfully demonstrated secure key distribution using quantum principles. The simulation showed that any eavesdropping attempt introduces detectable errors, ensuring communication security.

Assignment 10: Quantum Walks

Title

Implementation and Analysis of Quantum Walks on Graphs

Aim

To implement a quantum walk, simulate its behavior on a 1D line, and compare it with classical random walks.

Theory

1. Classical Random Walk

A particle moves randomly:

- Left or right with equal probability
 - Distribution becomes Gaussian over time
-

2. Quantum Walk

Quantum version uses:

- Superposition
- Interference

State:

$$|\psi\rangle = \sum \alpha_i |i\rangle$$

3. Types of Quantum Walks

- Discrete-time quantum walk
 - Continuous-time quantum walk
-

4. Components

- Coin operator (creates superposition)
 - Shift operator (moves particle)
-

5. Key Difference from Classical Walk

Feature	Classical	Quantum
Behavior	Random	Interference-based
Spread	Slow	Faster
Distribution	Gaussian	Non-Gaussian

6. Probability Distribution

- Classical → centered distribution
 - Quantum → spread with peaks
-

7. Applications

- Search algorithms
 - Graph traversal
 - Quantum simulations
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Conclusion

Quantum walks demonstrated faster spreading compared to classical random walks due to superposition and interference. This highlights their importance in quantum algorithms, especially in search and optimization problems.